

IBM BOB 2.0 HACKATHON

CurbCut

Accessibility review that proves its fixes.

curbcut.eziedutech.dev · github.com/eziedutech/curbcut

THE PROBLEM

This quiz form passed code review.

```
img, image  
textbox, unlabeled  
button, unlabeled  
(the Sign up button is missing)
```

What a screen reader user hears on it.

AI coding assistants now write more and more interface code, yet their output still overlooks accessibility. Most reviewers are not WCAG experts, so it passes review. Speed has improved; inclusion has not kept pace.

Four steps in every pull request

1. Measure

A deterministic engine records facts: axe-core, contrast, keyboard traps, focus.

2. Review

IBM Bob reviews with four subagents, one per WCAG principle.

3. Fix

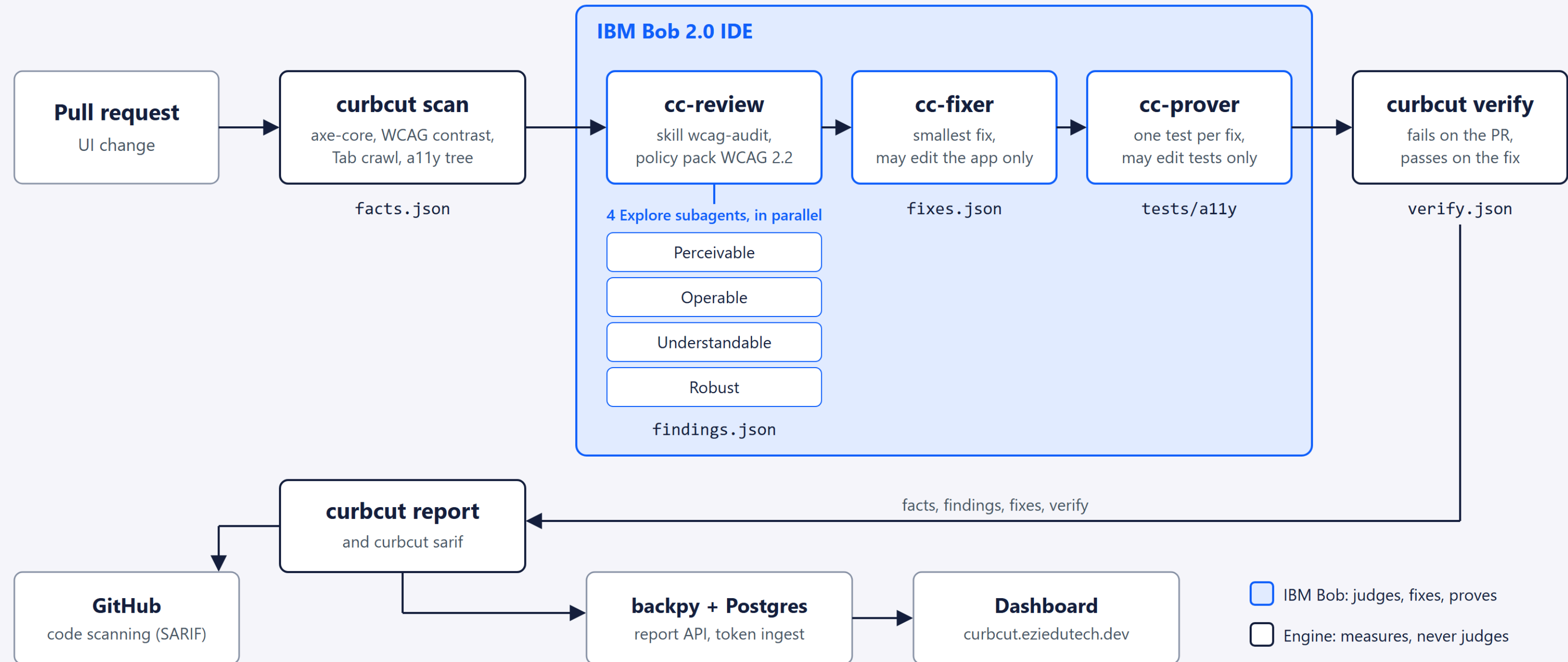
IBM Bob makes the smallest change per finding.

4. Prove

IBM Bob writes a test that fails before the fix and passes after.

Architecture

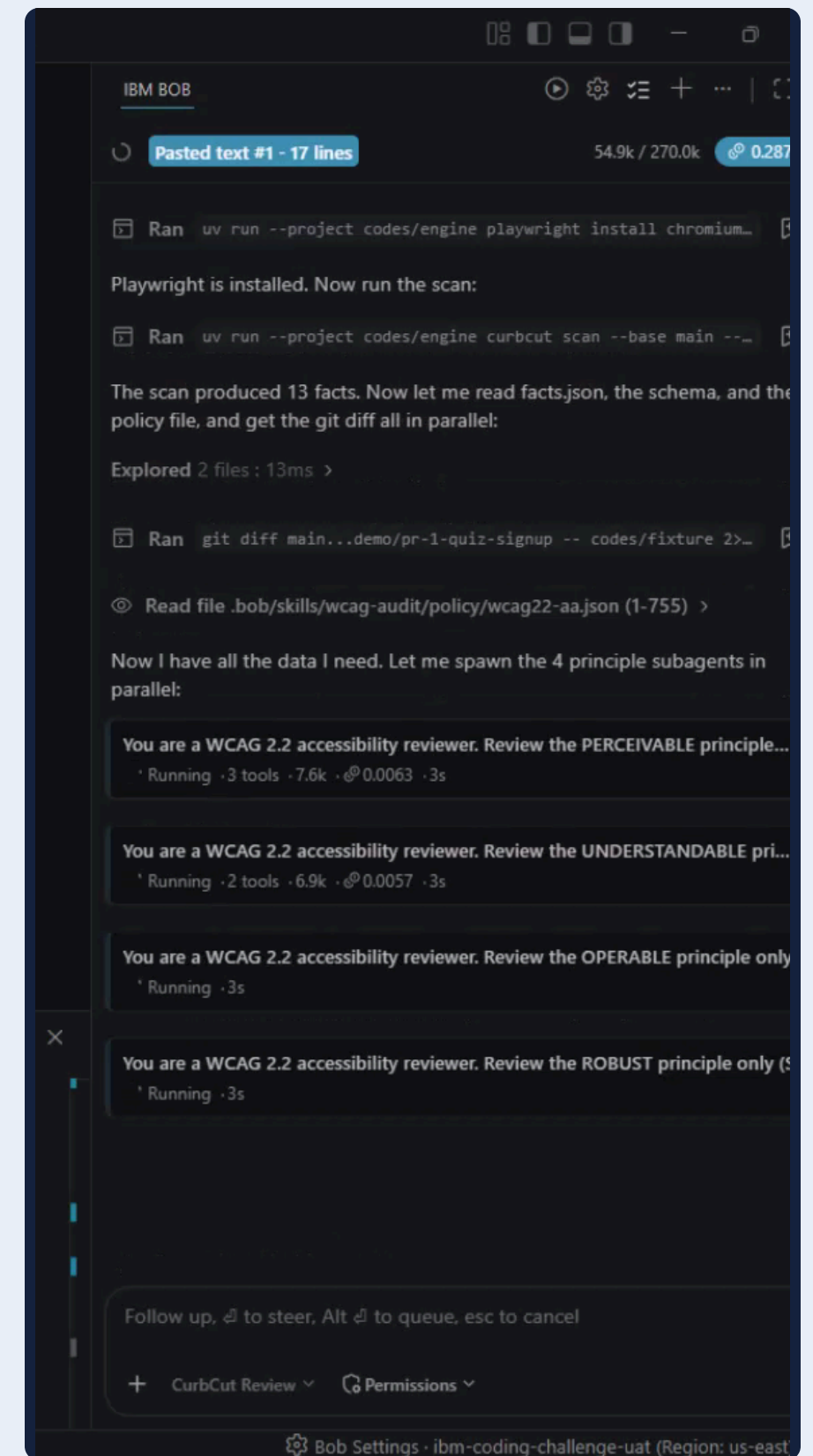
CurbCut: accessibility review that proves its fixes



HOW IBM BOB IS USED

Built from Bob's own building blocks

- **Skill wcag-audit:** Bob read WCAG 2.2 and wrote a policy pack of all 55 A and AA criteria.
- **Three custom modes:** review, fixer, prover. Edit rights limited by fileRegex.
- **Four Explore subagents in parallel:** read-only reviewers, one per principle.



Every finding says how it is known

Tier	Meaning
Proven	Measured by the engine. Cites a fact.
Flagged	IBM Bob's judgment, with a reason. Can be wrong.
Out of reach	Cannot be checked this way. Always listed.
Verified fix	Test fails on the pull request, passes on the fix.

Before and after, as a screen reader hears it

 CurbCut

OverviewEvidenceAbout

[Overview](#) /PR-1

PR-1: demo/pr-1-quiz-signup

eziedutech/curbcut, main at 736bb7f to demo/pr-1-quiz-signup at 250923a

Proven ⓘ
12

Flagged ⓘ
11

Out of reach ⓘ
13

Not scanned ⓘ
0

Verified fixes ⓘ
23/23

Screen reader narration ⓘ

Before the fix

link, Skip to main content

link, OpenClass

list

link, Courses

link, Biology basics

link, Quiz sign-up

link, Biology basics

heading, Sign up for the Biology quiz, level 1

img, image

textbox, unlabeled

textbox, Display name

textbox, Course code

combobox, Course

textbox, Date

radio, Starter, checked

radio, Core

After the fix

link, Skip to main content

link, OpenClass

list

link, Courses

link, Biology basics

link, Quiz sign-up

link, Biology basics

heading, Sign up for the Biology quiz, level 1

img, Cell Explorer badge

textbox, Email

textbox, Display name

textbox, Course code

combobox, Course

textbox, Date

radio, Starter, checked

radio, Core

23 / 23

fixes on PR-1 proven by a test that fails before the fix and passes after.

After the fix, the engine measures 0 of the 13 violations it found.

BENCHMARK

Planted issues found, same element and criterion

Pull request	axe-core alone	CurbCut with Bob	Decoys flagged
PR-1, run 1	8 of 18	16 of 18	0 of 6
PR-1, run 2	8 of 18	14 of 18	0 of 6
PR-2	6 of 16	14 of 16	0 of 6

Bob finds issues no rule can: unclear alt text, vague links, errors shown by colour only.

Demo app with planted issues. The two PR-1 runs differ only in judgment findings: run-to-run variation.

WHY IT IS DIFFERENT

Beyond a linter, beyond a general review

Today	CurbCut adds
Rule-based scanners	Judgment on what rules cannot see, clearly labelled
Manual audits	A review in every pull request, not once a year
General AI code review	A WCAG policy pack, evidence tiers, fixes proven by tests

For teams that must not leave anyone out

Who

Education, public service and non-profit product teams.

Why

Issues caught in review cost less than issues found after release.

Model

Free for open source and non-profits. Paid per private repository. Plans for schools.

WHAT IT DOES NOT CLAIM

Honest limits

- It does not certify WCAG conformance. Testing with disabled people cannot be replaced.
- It does not judge real screen reader experience, cognitive load or caption quality.
- Flagged findings are IBM Bob's judgment and can be wrong.
- The benchmark uses a demo app with issues planted by our team.

NEXT

From the IDE into every pull request

- Run the review when a pull request opens.
- A status check that blocks merge on proven critical findings.
- Explanations in Indonesian for local education teams.